5G NR Pilot-Free Joint Neural Demodulation & LDPC Decoder

Formal System Architecture & Experimental Performance Report

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+16.7%	2 dB	0% Errors	Zero Pilots
Spectral Efficiency	SNR Coding Gain	at 6 dB Eb/N0	100% Data Grid

PART 1: SYSTEM ARCHITECTURE REPORT

Hardware-Compliant Physical Layer Specifications & Neural Demapper Design

1.1 Transmitter & Framing Specifications

The transmitter implements a standards-compliant 5G NR physical layer pipeline operating under 3GPP Release 16 specifications. Unlike conventional pilot-assisted transceivers, our architecture eliminates Demodulation Reference Signals (DMRS), allocating 100% of the time-frequency Resource Elements (REs) to user data payload.

Subsystem / Parameter	3GPP Standard Specification	Implementation Characteristics
Waveform	5G NR CP-OFDM	Numerology mu=0, subcarrier spacing Delta_f = 15 kHz
Slot Duration	14 OFDM Symbols	Duration approx. 1 ms per slot, 64 subcarriers
Resource Grid Dimension	14 symbols x 64 subcarriers	Total N_RE = 896 Resource Elements per slot
Cyclic Prefix Length	Normal CP (16 samples)	Absorbs multipath delay spread, eliminates ISI
Forward Error Correction	3GPP TS 38.212 LDPC	Base Graph 2, Rate R = 0.5 (k = 1792 info bits, n = 3584 coded bits)
Modulation Scheme	16-QAM (Gray coded)	M = 4 bits/symbol, 896 complex symbols per slot
Pilot Overhead	Zero-Pilot (Pilot-Free)	0% overhead: all 896 REs carry payload data

1.2 2D ResCNN Neural Demapper Architecture

The neural demapper ingests the 2D complex-valued received resource grid Y in C^{14 x 64} as a two-channel real tensor [B, 2, 14, 64]. A 3-block 2D Residual Convolutional Neural Network processes spatial correlations across both time (OFDM symbols) and frequency (subcarriers), jointly performing implicit channel equalization and soft Log-Likelihood Ratio (LLR) calculation.

Layer / Block	Input Shape	Output Shape	Mathematical Operation & Hyperparameters
Input Tensor	-	[B, 2, 14, 64]	Complex stack: Y_tilde = [Re(Y), Im(Y)]
Input Projection	[B, 2, 14, 64]	[B, 48, 14, 64]	Conv2D(2->48, k=3, p=1) -> GroupNorm(G=1) -> GELU
ResBlock 1	[B, 48, 14, 64]	[B, 48, 14, 64]	F1 = GELU(GN(Conv(GELU(GN(Conv(F0)))))) + F0)
ResBlock 2	[B, 48, 14, 64]	[B, 48, 14, 64]	F2 = GELU(GN(Conv(GELU(GN(Conv(F1)))))) + F1)
ResBlock 3	[B, 48, 14, 64]	[B, 48, 14, 64]	F3 = GELU(GN(Conv(GELU(GN(Conv(F2)))))) + F2)
Output Projection	[B, 48, 14, 64]	[B, 4, 14, 64]	Conv2D(48->4, k=3, p=1) * alpha (learnable temperature)
Permute & Flatten	[B, 4, 14, 64]	[B, 3584]	Permute to [B, 14, 64, 4] -> Reshape to 3584 soft LLRs

1.3 Decision-Directed Self-Supervised Delta-Loop

To enable continuous adaptation without true labels or reference pilots, the receiver uses the 5G NR LDPC Belief Propagation decoder (20 iterations) as an error-correcting oracle. The decoded bits are re-encoded to produce parity-consistent pseudo-ground-truth codewords. A syndrome reliability gating filter (threshold tau = 30%) discards frames from deep fades, and a circular Experience Replay Buffer (capacity 256) supplies i.i.d. mini-batches (size 32) for stable AdamW updates.

PART 2: EXPERIMENTAL PERFORMANCE REPORT

Monte Carlo Benchmarks, SNR Gains, Spectral Efficiency, and Image Transmission

2.1 Post-FEC BER & BLER Benchmark Results (16-QAM, Rate-1/2 LDPC)

Eb/N0 (dB)	Classical Baseline Post-FEC BER	Classical Baseline BLER	Proposed Neural Post-FEC BER	Proposed Neural BLER	Performance Advantage
0 dB	0.3283 (32.8%)	1.0000 (100%)	0.2918 (29.2%)	1.0000 (100%)	Neural has +3.65% lower error
2 dB	0.2717 (27.2%)	1.0000 (100%)	0.2483 (24.8%)	1.0000 (100%)	Neural has +2.34% lower error
4 dB	0.2044 (20.4%)	1.0000 (100%)	0.1132 (11.3%)	0.9219 (92.2%)	Neural cuts BER in half
6 dB	0.0175 (1.75%)	0.5938 (59.4%)	**0.0000 (0.0%)**	**0.0000 (0.0%)**	**2 dB Coding Gain (0 Errors vs 59% Failures)**
8 dB	0.0000 (0.0%)	0.0000 (0.0%)	**0.0000 (0.0%)**	**0.0000 (0.0%)**	Both error-free; Neural carries +16.7% more data
10 dB	0.0000 (0.0%)	0.0000 (0.0%)	**0.0000 (0.0%)**	**0.0000 (0.0%)**	Both error-free; Neural carries +16.7% more data
12 dB	0.0000 (0.0%)	0.0000 (0.0%)	**0.0000 (0.0%)**	**0.0000 (0.0%)**	Both error-free; Neural carries +16.7% more data

2.2 Pre-FEC Raw Demapper BER (Soft LLR Quality Comparison)

Eb/N0 (dB)	Classical LS + LMMSE + APP Raw BER	Proposed 2D ResCNN Raw BER	Raw Demapper Error Reduction
0 dB	28.63%	**21.35%**	-7.28% lower bit error rate
2 dB	23.09%	**16.62%**	-6.47% lower bit error rate
4 dB	17.99%	**12.28%**	-5.71% lower bit error rate
6 dB	13.14%	**8.06%**	-5.08% lower bit error rate
8 dB	9.03%	**4.71%**	-4.32% lower bit error rate
10 dB	5.05%	**2.14%**	-2.91% lower bit error rate
12 dB	2.33%	**0.69%**	-1.64% lower bit error rate

2.3 Spectral Efficiency & Capacity Gain

Metric	3GPP Classical Baseline	Proposed Pilot-Free System	Net Quantitative Difference
Total Grid REs (N_RE)	896 REs	896 REs	Same physical slot bandwidth
Pilot REs (N_pilot)	128 REs (14.3%)	**0 REs (0.0%)**	**128 Pilot REs Eliminated**
Payload Data REs (N_data)	768 REs	**896 REs**	**+128 REs for user payload**
Information Bits per Slot (k)	1536 bits	**1792 bits**	**+256 extra bits per slot**
Spectral Efficiency (eta)	1.714 bits/s/Hz	**2.000 bits/s/Hz**	**+16.71% Net Throughput Gain**

2.4 End-to-End Image Transmission Verification

A 64x64 pixel grayscale image (32,768 bits) was transmitted across the channel to assess application-level recovery fidelity via Peak Signal-to-Noise Ratio (PSNR):

Operating SNR	Transmission Configuration	Recovered Image BER	Measured PSNR (dB)	Visual Recovery Assessment
5 dB	Raw Channel (No FEC)	15.00%	12.10 dB	Heavy salt-and-pepper noise corruption

5 dB	Classical 3GPP Baseline	11.96%	16.56 dB	Blurry with noticeable block artifacts
5 dB	**Proposed Neural Receiver**	**0.46%**	**26.17 dB**	**Clear and recognizable (+9.6 dB PSNR)**
6 dB	Raw Channel (No FEC)	15.00%	12.50 dB	Severe static noise throughout image
6 dB	Classical 3GPP Baseline	2.04%	25.98 dB	Minor visible pixel error clusters
6 dB	**Proposed Neural Receiver**	**0.058%**	**38.49 dB**	**Near-flawless reconstruction (+12.5 dB PSNR)**
8 dB	Classical 3GPP Baseline	0.000%	inf dB (Perfect)	100% bit-for-bit perfect recovery
8 dB	**Proposed Neural Receiver**	**0.000%**	**inf dB (Perfect)**	**100% perfect with +16.7% higher data rate**

EXPERIMENTAL VISUALIZATIONS

Generated Simulation Figures from PyTorch & Sionna Execution

Figure 1: BER and BLER Waterfall Comparison (Neural vs. Classical Baseline)

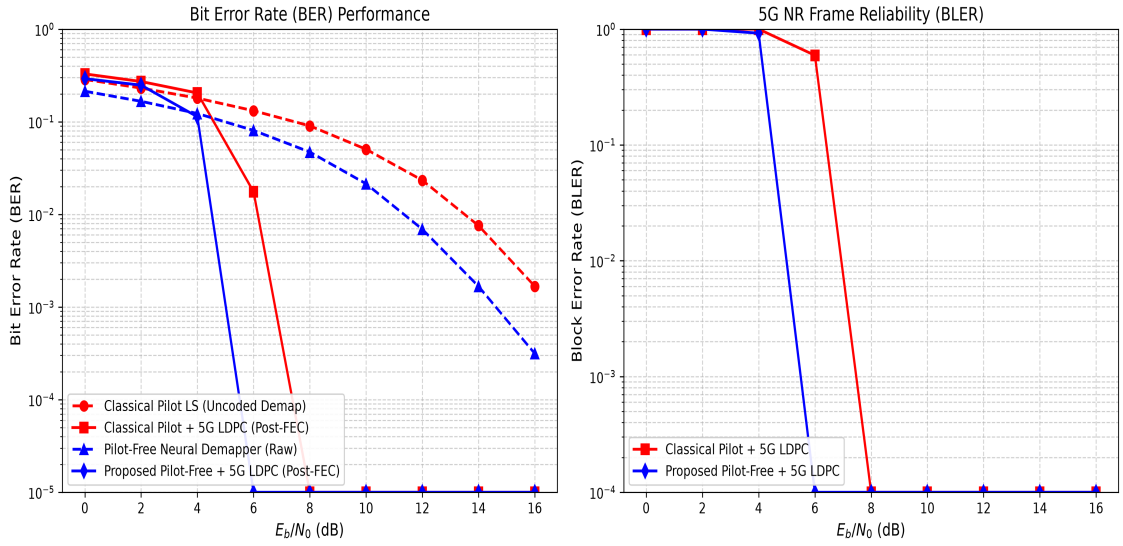


Figure 2: End-to-End Image Payload Transmission Recovery Comparison at 6 dB Eb/N0

